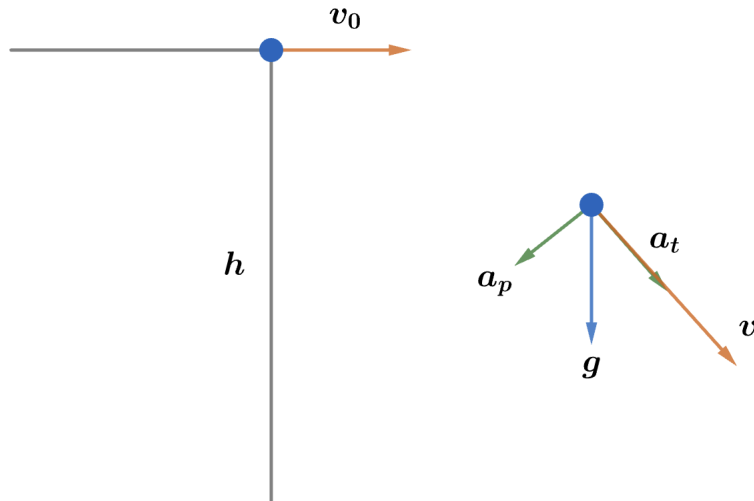


2007 F=ma Exam: Problem 17

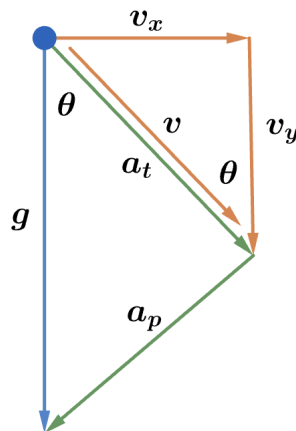
Kevin S. Huang



Since the object is undergoing projectile motion, its acceleration is always directed downward with magnitude g . At any point along the trajectory, we can split g into a tangential component parallel to the velocity and a perpendicular component. The direction of the velocity vector can be determined from its horizontal and vertical components. From kinematics, we know

$$v_x(t) = v_0$$

$$v_y(t) = gt$$



This can be related to the acceleration components by similar triangles. We have

$$\tan \theta = \frac{a_p}{a_t} = \frac{v_x}{v_y} = \frac{v_0}{gt}$$

In our case, we want

$$\begin{aligned}\frac{1}{2} &= \frac{a_p}{a_t} = \frac{v_0}{gt} \\ t &= \frac{2v_0}{g} = \frac{2(10 \text{ m/s})}{10 \text{ m/s}^2} = 2 \text{ s}\end{aligned}$$

In this time, the object falls a distance

$$y = \frac{1}{2}gt^2 = \frac{1}{2}(10 \text{ m/s}^2)(2 \text{ s})^2 = 20 \text{ m}$$

Since $y < h$, the building is high enough for this to occur so the answer is A.